# Year 10 Mathematics

## Graphs 1

3.1 Gradient of a Straight Line

# Gradient of a Straight Line

STARTER · 5 MIN

**Date:** _____ **Title:** Gradient of a Straight Line

**LEARNING OBJECTIVES**

- Find the gradient of a straight line from a graph
- Find the gradient of a line through two points
- Use gradients to solve problems

**STARTER** *Answer in your book. No calculators.*

1. Work out $12 \div 4$
2. Work out $-8 \div 2$
3. Work out $(9 - 3) \div (5 - 2)$
4. Work out $(2 - 10) \div (6 - 2)$

# Starter — Answers

STARTER

*Tick or correct in green pen*

1. $12 \div 4 =$ **3**
2. $-8 \div 2 =$ **$-4$**
3. $6 \div 3 =$ **2**
4. $-8 \div 4 =$ **$-2$**

# Success Criteria

SUCCESS CRITERIA

*By the end of this lesson:*

| | | |
|---|---|---|
| BRONZE | I can find the gradient of a straight line from a graph | ☐ |
| SILVER | I can find the gradient of a line through two points | ☐ |
| GOLD | I can use gradients to solve problems | ☐ |

*Tick each box in the plenary when you can do it on your own.*

# Introduction — Gradient

INTRODUCTION · 3 MIN

The gradient tells you how steep a line is.

## Rise over run

Gradient = change in y ÷ change in x

$$m = \frac{\text{rise}}{\text{run}}$$

## Positive

The line goes up from left to right.

$m > 0$

## Negative

The line goes down from left to right.

$m < 0$

## Two points

Subtract the y's, then subtract the x's.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

# Examples 1 and 2 — Gradient from a Graph

EXAMPLES · 5 MIN

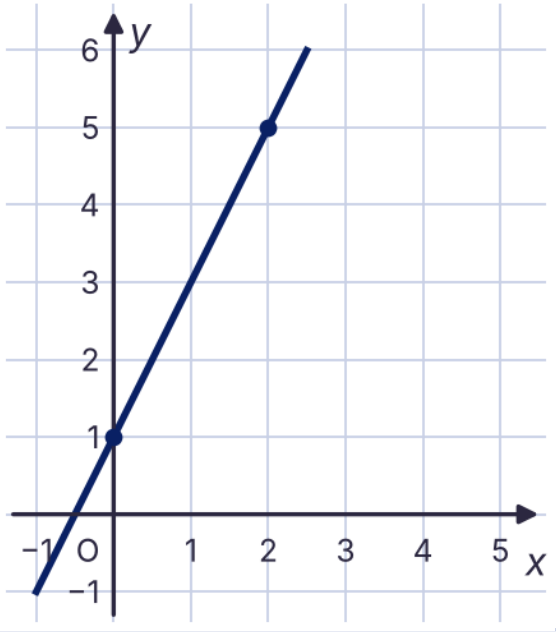
BRONZE

*Watch and copy into your book*

**Hint** $m = \frac{\text{change in y}}{\text{change in x}}$

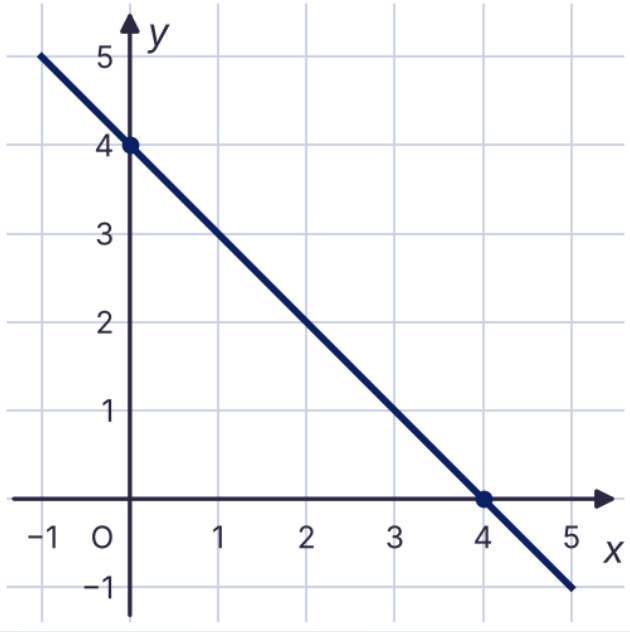
## Example 1

Find the gradient of this line.




## Example 2

Find the gradient of this line.



# Examples 1 and 2 — Corrections

CORRECTIONS

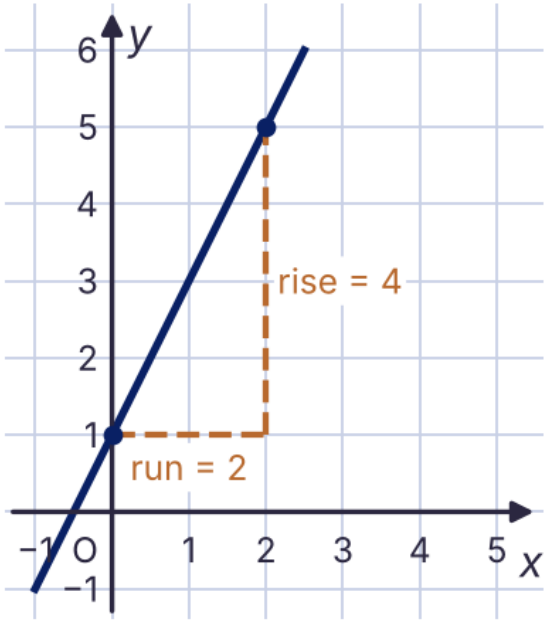
BRONZE

*Check your copy and correct in green pen*

## Example 1

$$m = \frac{\text{rise}}{\text{run}} = \frac{4}{2}$$

**m = 2**

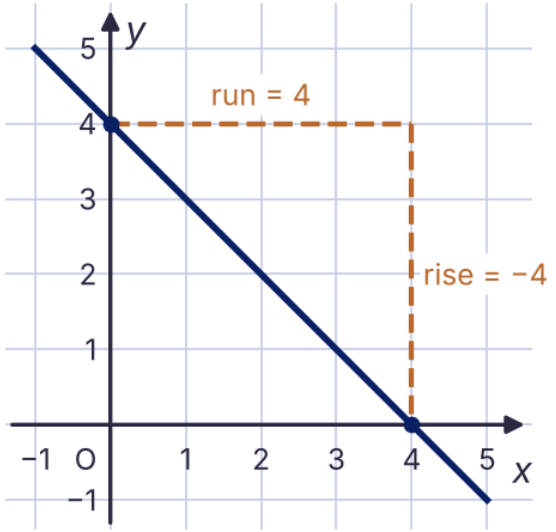



Tip: pick two points where the line crosses grid corners.

## Example 2

$$m = \frac{\text{rise}}{\text{run}} = \frac{-4}{4}$$

**m = −1**




Tip: the line goes down, so the gradient is negative.

# Task 1

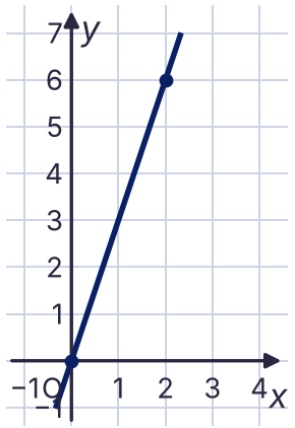
PRACTICE · 9 MIN

BRONZE

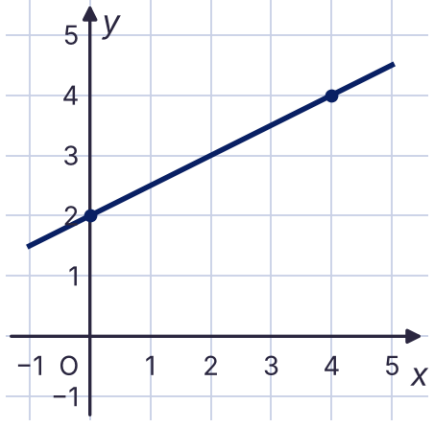
*Find the gradient of each line*

**Hint** $m = \dfrac{\text{change in y}}{\text{change in x}}$

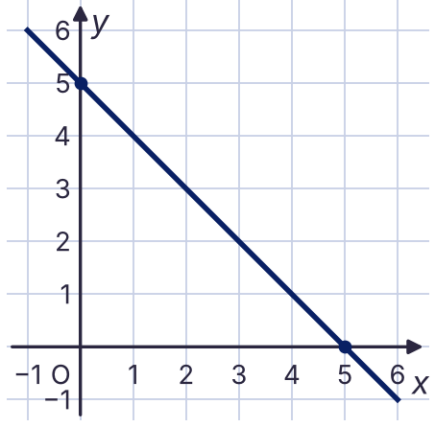
**1**



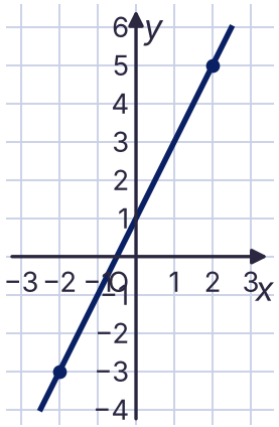

**2**



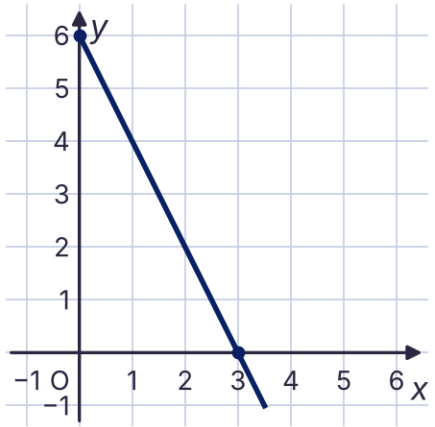

**3**



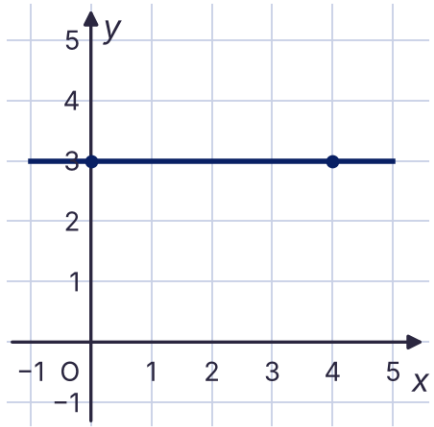

**4**




**5**




★



# Task 1 — Answers

CORRECTIONS

BRONZE

*Tick or correct in green pen*

1 $m = \frac{6}{2} = \mathbf{3}$

2 $m = \frac{2}{4} = \mathbf{\frac{1}{2}}$

3 $m = \frac{-5}{5} = \mathbf{-1}$

4 $m = \frac{8}{4} = \mathbf{2}$

5 $m = \frac{-6}{3} = \mathbf{-2}$

★ $m = \frac{0}{4} = \mathbf{0}$

# Examples 3 and 4 — Gradient Between Two Points

EXAMPLES · 5 MIN

SILVER

*Watch and copy into your book*

**Hint** $m = \frac{y_2 - y_1}{x_2 - x_1}$

## Example 3

Find the gradient of the line through

(1, 3) and (4, 12)

## Example 4

Find the gradient of the line through

(−2, 5) and (2, −3)

# Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

*Check your copy and correct in green pen*

## Example 3

$$m = \frac{12 - 3}{4 - 1}$$

$$= \frac{9}{3}$$

**m = 3**

Tip: always subtract in the same order.

## Example 4

$$m = \frac{-3 - 5}{2 - (-2)}$$

$$= \frac{-8}{4}$$

**m = −2**

Tip: take care with negative numbers.

# Task 2

PRACTICE · 9 MIN

SILVER

*Find the gradient of the line through each pair of points*

**Hint** $m = \frac{y_2 - y_1}{x_2 - x_1}$

1. (2, 1) and (5, 7)
2. (0, −3) and (2, 9)
3. (1, 6) and (3, 2)
4. (−1, 4) and (3, 6)
5. (−2, 7) and (4, −2)

# Task 2 — Answers

CORRECTIONS

SILVER

*Tick or correct in green pen*

**1** $m = \frac{7-1}{5-2} = \frac{6}{3} = \mathbf{2}$

**2** $m = \frac{9-(-3)}{2-0} = \frac{12}{2} = \mathbf{6}$

**3** $m = \frac{2-6}{3-1} = \frac{-4}{2} = \mathbf{-2}$

**4** $m = \frac{6-4}{3-(-1)} = \frac{2}{4} = \mathbf{\frac{1}{2}}$

**5** $m = \frac{-2-7}{4-(-2)} = \frac{-9}{6} = \mathbf{-\frac{3}{2}}$

# Examples 5 and 6 — Gradient Problems

EXAMPLES · 5 MIN

GOLD

*Exam-style: show all your working*

**Hint** $m = \dfrac{y_2 - y_1}{x_2 - x_1}$

## Example 5

(3 marks)

A line passes through (2, k) and (5, 11).

The gradient of the line is 3.

Find the value of k.

## Example 6

(3 marks)

A(−1, 2), B(3, 10) and C(5, 14)

Show that A, B and C lie on the same straight line.

# Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

*Check your copy and correct in green pen*

## Example 5

$$\frac{11-k}{5-2}=3$$

$$11-k=9$$

$$\mathbf{k=2}$$

Tip: write the gradient formula, then solve.

## Example 6

$$AB=\frac{10-2}{3-(-1)}=\frac{8}{4}=2$$

$$BC=\frac{14-10}{5-3}=\frac{4}{2}=2$$

Same gradient and B is shared,

**so A, B and C are on one line**

Tip: equal gradients and a shared point mean one straight line.

# Task 3

PRACTICE · 9 MIN

GOLD

*Exam-style: show all your working*

**Hint** $m = \dfrac{y_2 - y_1}{x_2 - x_1}$

1. Find the gradient of the line through (−4, −1) and (2, 2) (2 marks)

2. A line through (1, 4) and (3, k) has gradient 5. Find k. (2 marks)

3. A line through (p, 3) and (6, 11) has gradient 2. Find p. (3 marks)

4. A ramp rises 1.2 m over a horizontal distance of 4.8 m. Find the gradient of the ramp. (2 marks)

5. Show that P(0, 1), Q(2, 7) and R(4, 13) lie on a straight line. (3 marks)

# Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1 $m = \frac{2-(-1)}{2-(-4)} = \frac{3}{6} = \mathbf{\frac{1}{2}}$

2 $\frac{k-4}{2} = 5, \quad k-4 = 10, \quad k = \mathbf{14}$

3 $\frac{8}{6-p} = 2, \quad 6-p = 4, \quad p = \mathbf{2}$

4 $m = \frac{1.2}{4.8} = \mathbf{0.25}$

5 $PQ = \frac{6}{2} = 3, \quad QR = \frac{6}{2} = 3,$ **Same gradient, so a straight line**

# Plenary — Exit Ticket

PLENARY · 5 MIN

*Answer in your book. Try all three.*

**BRONZE** Find the gradient of the line through (0, 0) and (3, 12)

**SILVER** Find the gradient of the line through (−2, 3) and (2, −5)

**GOLD** A line through (3, k) and (7, 15) has gradient 2. Find k.

**Reflect:** Look back at the learning objectives. Which ones can you do now?

# Plenary — Answers

PLENARY

*Tick or correct in green pen*

**BRONZE** $m = \frac{12}{3} = \mathbf{4}$

**SILVER** $m = \frac{-5-3}{2-(-2)} = \frac{-8}{4} = \mathbf{-2}$

**GOLD** $\frac{15-k}{4} = 2, \quad 15-k = 8, \quad k = \mathbf{7}$

# Exam Practice — Gradient from a Graph

EXAM PRACTICE · 10 MIN

BRONZE

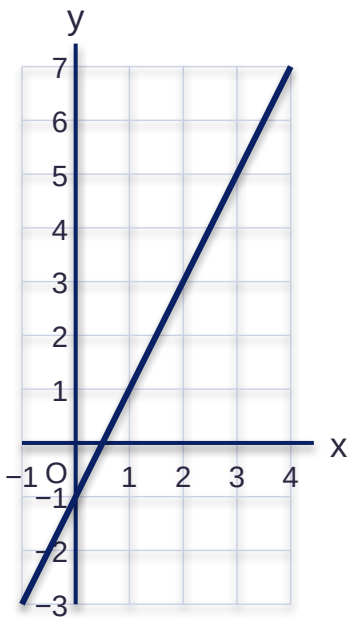
*IGCSE exam-style: show all your working*

**Hint** $m = \frac{\text{change in y}}{\text{change in x}}$

## Question 1

(2 marks)

Find the gradient of the straight line drawn on the grid.

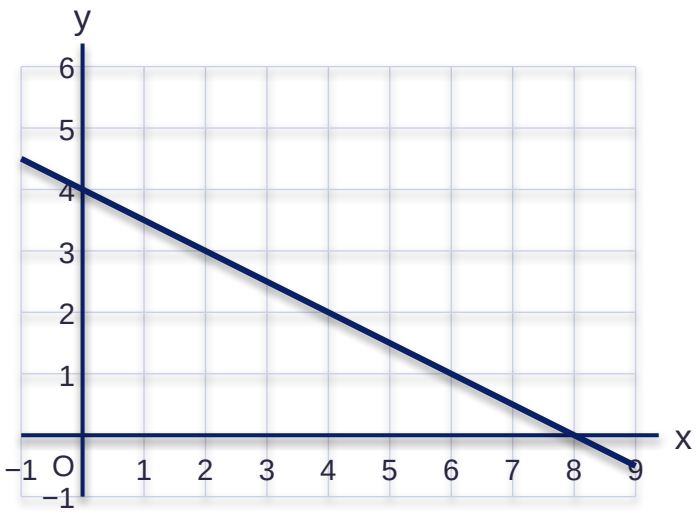



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## Question 2

(2 marks)

Find the gradient of the straight line drawn on the grid.




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# Exam Practice — Gradient Between Two Points

EXAM PRACTICE · 10 MIN

SILVER

*IGCSE exam-style: show all your working*

**Hint** $m = \frac{y_2 - y_1}{x_2 - x_1}$

## Question 3

(2 marks)

A is the point (2, 5).

B is the point (6, 17).

Work out the gradient of the line AB.

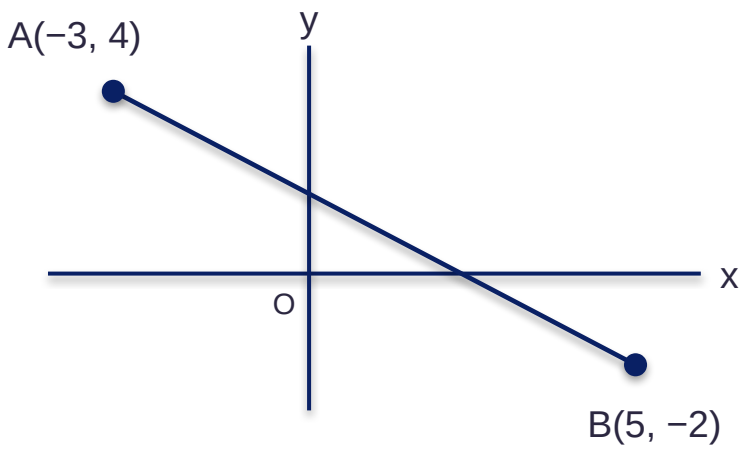
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## Question 4

(2 marks)

Work out the gradient of the line AB.

*Diagram NOT accurately drawn*




.....

# Exam Practice — Problems with Gradients

EXAM PRACTICE · 10 MIN

GOLD

*IGCSE exam-style: show all your working*

**Hint** $m = \dfrac{y_2 - y_1}{x_2 - x_1}$

## Question 5

(3 marks)

A straight line passes through the points (1, 3) and (4, k).

The gradient of the line is 2.

Work out the value of k.

k = .....

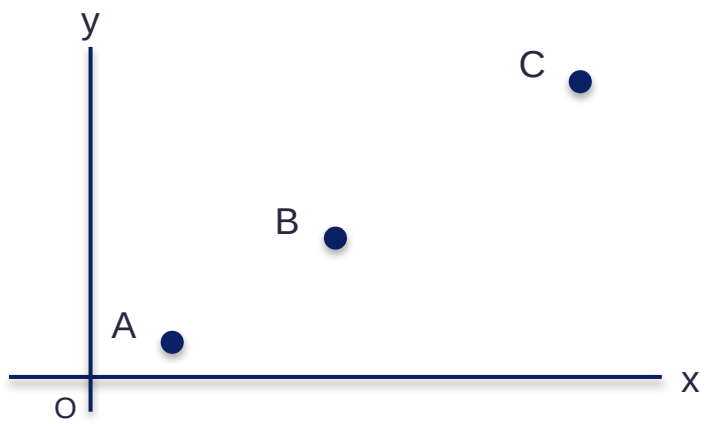
## Question 6

(3 marks)

A is (1, 2), B is (3, 8) and C is (6, 17).

Show that A, B and C lie on the same straight line.

*Diagram NOT accurately drawn*



# Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 1

e.g. (0, −1) to (3, 5): rise 6, run 3 — M1

**Gradient = 2** — A1

Tip: use two points on grid corners.

## Question 2

e.g. (0, 4) to (4, 2): rise −2, run 4 — M1

**Gradient = $-\frac{1}{2}$** — A1

Tip: down to the right ⇒ negative.

# Exam Practice — Mark Scheme

CORRECTIONS

SILVER

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 3

$(17 - 5) \div (6 - 2)$ — M1

**$= 12 \div 4 = 3$** — A1

Tip: change in y ÷ change in x.

## Question 4

$(-2 - 4) \div (5 - (-3))$ — M1

$= -6 \div 8$

**$= -\frac{3}{4}$** — A1

Tip: subtracting a negative: 5 − (−3) = 8.

# Exam Practice — Mark Scheme

CORRECTIONS

GOLD

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 5

| | |
|---|---|
| (k − 3) ÷ (4 − 1) = 2 | M1 |
| k − 3 = 6 | M1 |
| **k = 9** | A1 |

Tip: write the gradient formula, then solve.

## Question 6

| | |
|---|---|
| AB: (8 − 2) ÷ (3 − 1) = 3 | M1 |
| BC: (17 − 8) ÷ (6 − 3) = 3 | M1 |
| **Same gradient, shared point B** | A1 |

Tip: equal gradients AND a common point.